

Tracked QEM Algorithm: Adding Temporal Consistency to Dynamic Mesh Simplification Based on Mesh Registration

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Abstract The evolution of 3D multi-media technology has spurred the need for more effective 3D video storage and transmission methods. Most of the current standardizing 3D volumetric video coding methods in the mesh decimation stage are still limited: the difference of geometric structure in each simplified frame hinders optimal compression. The proposed Tracked QEM Algorithm effectively tracks mesh models across successive frames, offering a tailored solution for dynamic meshes in 3D volumetric videos.

The Tracked QEM Algorithm ensures that the simplified results have better topological consistency and spatial smoothness between consecutive frames than the original QEM algorithm. The evaluation results based on temporal consistency show that the proposed approach is superior to the conventional mesh simplification. The smoother simplified results with similar topology delineate the discontinuous structural information between frames. As a novel pre-processing approach to 3D video compression, this proposal has the potential to improve the compression rate.

Key words: Multi-media Technology, Volumetric Video, Computer Graphic, Mesh Simplification, Mesh Registration, QEM Algorithm

1. Introduction

Over the past few years, Augmented and Virtual Reality (AR/VR) technologies¹⁾ developed rapidly²⁾. Central to these advanced multi-media technologies, 3D volumetric video³⁾ is pivotal in accurately recreating 3D objects and scenes, gaining significant attention⁴⁾. With the ongoing actions in standardizing 3D volumetric video coding⁵⁾, there is a critical need to enhance storage and transmission efficiency for this particular 3D video format.

Similar to 2D video compression⁶⁾, 3D volumetric video encoding follows a multi-step pipeline⁷⁾, a concise summary can be depicted in Fig.1. The encoding progress begins by taking a series of raw frames as input, transforming them into a continuous bitstream for efficient storage and transmission⁸⁾. As the red part shown in this figure, an essential component of this pipeline is mesh decimation, which reduces the num-

ber of faces in the input mesh to diminish the amount of data. Given that dynamic meshes in 3D videos consist of a series of static mesh frames, current encoding standards⁹⁾ rely on static mesh simplification methods.

Represented by the Quadratic Error Metric (QEM) algorithm¹⁰⁾, these methods can efficiently simplify the individual static mesh frame¹¹⁾. As shown in Fig.2, the simplification algorithm uses the mesh of each frame as an input to generate corresponding simplified results. However, due to less consideration of the inter-frame continuity, the difference between adjacent frames may be amplified in the simplification process. The simplified frames may differ in appearance and topology, which can be called inconsistency of the mesh decimation step.

Since the discontinuous structural information will lead to higher encoding expenses in bitstream phase¹²⁾, ensuring the consistency of frame sequences as much as possible in pre-processing stages becomes an essential issue in 3D video coding. A novel mesh decimation method is proposed to solve this problem. This method aims to add temporal consistency via the introduction of a reference frame.

As shown in Fig.3, considering the dynamic mesh consists of static mesh frames, the proposed Tracked

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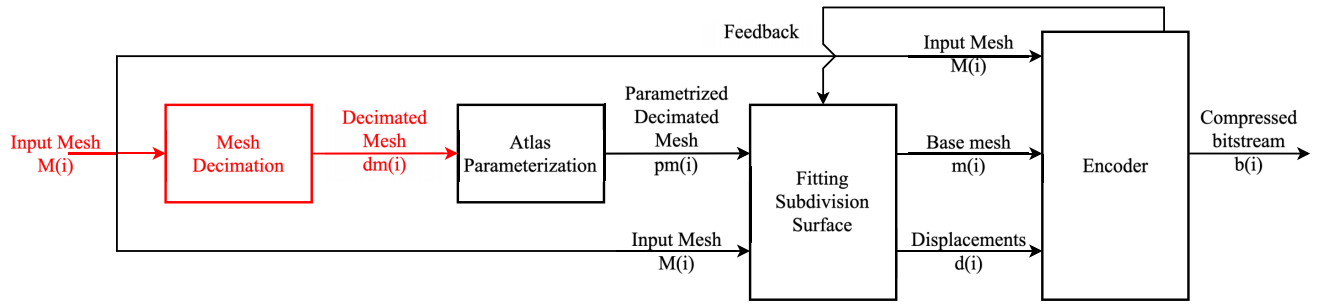


Fig. 1 A typical encoder pipeline for 3D Volumetric video

QEM algorithm tracks registered¹³⁾ mesh model from the previous frame to drastically boost the temporal consistency in dynamic mesh simplification.

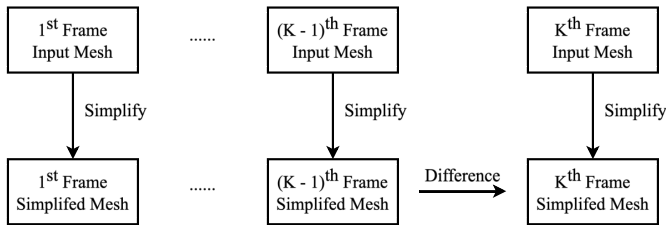


Fig. 2 Current mesh decimation in 3D video coding on individual frames

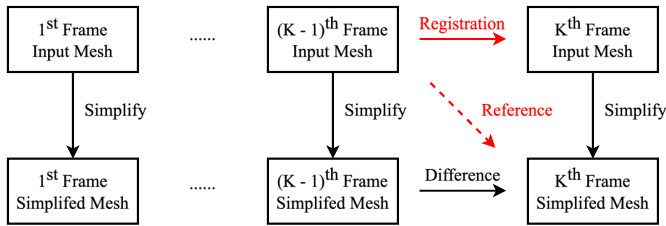


Fig. 3 Proposed dynamic mesh simplification with inter-frame tracking

Experiments are also carried out to compare the effectiveness in dynamic mesh decimation task between the proposed Tracked QEM Algorithm and previous Original QEM Algorithm. The result reveals that the proposal significantly improves the temporal consistency of the traditional approach. As a pre-processing step of 3D video coding, preparing a smoother, more coherent 3D frame sequence helps maintain the consistent topology of input meshes. This merit shows the potential of the Tracked QEM Algorithm to advance the related 3D volumetric video compression research in the future.

The main contributions of the proposed method in this paper are as follows:

- Innovatively introduces 3D mesh registration to explore the possibility of considering temporal consistency in mesh simplification tasks.

- Proposes a Tracked QEM Algorithm suitable for dynamic mesh in 3D volumetric video.

- Redesigns inter-frame collapsed cost function and optimal strategy, enabling QEM Algorithm to handle reference input.

2. Related work

2.1 Mesh Registration



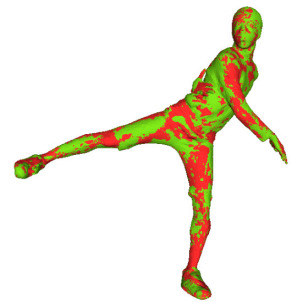
(a) Tracked mesh



(b) Reference mesh



(c) Rigid registration



(d) Non-Rigid registration

Fig. 4 Rigid & Non-Rigid Registration

The main work in this paper is a proposal of a novel mesh simplification method that innovatively integrates the principles of mesh registration. Mesh registration is a specific 3D registration in computer vision and computer graphics, which is a beneficial task that focuses on tracking and aligning distinct 3D models into a typical coordinate system¹³⁾. The registration has evolved

from rigid to non-rigid approaches that account for variations and distortions¹⁴). As shown in Fig.4, mesh registration can be divided into rigid and non-rigid registration.

Rigid Registration: Rigid Registration aligns two 3D image by applying an Euclidean transformations¹⁵, many related research has also made progress in various 3D image formats¹⁶). The rigid mesh registration evolved from approaches for point set matching^{17–19} from point cloud²⁰), the representative methods^{21–24} with many variants²⁵) is Iterative Closest Point (ICP)²⁶). As shown in Eq.(1), for each corresponding point pair p_i^c and p_j^r , ICP constructs the distance function between two point sets $\mathcal{P}^c = \{p_{1:N^c}^c\}$ and $\mathcal{P}^r = \{p_{1:N^r}^r\}$ with the inherent noise Ω_{ij} from 3D sensor, then iterates the minimum value of affine transformation matrix T via least square method to get the most suitable transformation matrix T^* ²⁷):

$$T^* = \arg \min_T \sum_c (p_i^c - T \oplus p_j^r)^T \Omega_{ij} (p_i^c - T \oplus p_j^r) \quad (1)$$

Non-Rigid Registration: In the context of non-rigid registration, it is imperative to account for the multifaceted nature of transformations that surpass the simplicity of rigid body movements²⁸). Embedded deformation²⁹) inspired from modified ICP algorithm^{30–32}) is a representative approach³³). As stated in Eq.(2), Like rigid registration, the early methods, aimed to generate the deformation field $\mathbf{X}$ with a transformation point set $\mathbf{G} = \{\mathbf{g}_j | \mathbf{g}_{1:N^x}\}$ from several transformation $\mathbf{X}_j$ with affine $\mathbf{R}_j$ and displace $\mathbf{t}_j$ matrices. The relationship between each target point $\hat{\mathbf{v}}_i$ and transformed source $\mathbf{v}_i$ can be described by several neighbour transform $\mathbf{X}_j$ weighted by w_j .

$$\hat{\mathbf{v}}_i = \sum_{j=1}^m w_j (\mathbf{v}_i) [\mathbf{R}_j (\mathbf{v}_i - \mathbf{g}_j) + \mathbf{g}_j + \mathbf{t}_j], \quad (2)$$

$$\mathbf{X}_j = (\mathbf{R}_j, \mathbf{t}_j), \mathbf{X}_j \in \mathbf{X}$$

Considering generality and usability, the robust iterative optimization-based methods^{34–37}) are more suitable for combining with upstream or downstream tasks. The approach integrated into proposed method is the Fast-RNRR³⁸), which constructed the optimal target $\mathbf{X}^*$ from cost function $E(\mathbf{X})$ through three sub-items in Eq.(3): alignment item $E_{\text{align}}(\mathbf{X})$ to penalizes the deviation between aligned $\hat{\mathbf{v}}_i$ and input $\mathbf{u}_{\rho(i)}$, regularization

item $E_{\text{reg}}(\mathbf{X})$ to avoid the over-fitting of deformation $\mathbf{D}_{ij}$ with weight α , and rotation item $E_{\text{rot}}(\mathbf{X})$ to make the deviation $\mathbf{A}_i$ close to a projection transformation $\text{proj}_{\mathcal{R}}(\mathbf{A}_i)$ weighted by β .

$$\mathbf{X}^* = \min_{\mathbf{X}} E_{\text{align}}(\mathbf{X}) + \alpha E_{\text{reg}}(\mathbf{X}) + \beta E_{\text{rot}}(\mathbf{X}),$$

$$\begin{aligned} E_{\text{align}}(\mathbf{X}) &= \sum_{i=1}^n \psi_{\nu_a} (\|\hat{\mathbf{v}}_i - \mathbf{u}_{\rho(i)}\|), \\ E_{\text{reg}}(\mathbf{X}) &= \sum_{i=1}^r \sum_{\mathbf{p}_j \in \mathcal{N}(\mathbf{p}_i)} \psi_{\nu_r} (\|\mathbf{D}_{ij}\|), \\ E_{\text{rot}}(\mathbf{X}) &= \sum_{i=1}^r \|\mathbf{A}_i - \text{proj}_{\mathcal{R}}(\mathbf{A}_i)\|_F^2, \\ \psi_{\nu}(x) &= 1 - \exp\left(-\frac{x^2}{2\nu^2}\right) \end{aligned} \quad (3)$$

This scheme introduces the Welsch function $\psi_{\nu}(\cdot)$ ³⁹) to ensure the smoothness of the optimization. Fast-RNRR uses the Quasi-Newton solver combining MM and L-BFGS to improve optimization efficiency, which has rapid convergence speed without prior data or training.

2.2 Mesh Simplification



(a) Original input (b) Simplified result
Fig. 5 A visualized case for input and output of typical mesh simplification

The other part of the related work is structured around mesh simplification techniques as the basic knowledge, which helps to contextualize the QEM algorithm used in the proposal. Mesh Simplification is a crucial research direction in 3D computer graphics processing. As shown in Fig.5, it is a procedure to reduce the number of vertices, edges, and faces

in a mesh model⁴⁰⁾. Meanwhile, retain the original model's appearance and topological structure with numerous attempts⁴¹⁾. These researches⁴²⁾ emerged in the 1990s^{43–46)}, which aims to automatically generate different Levels of Detail (LOD) for 3D models⁴⁷⁾.

Early methods focused on the geometric error^{48–51)}, ensuring retaining the original appearance after simplification. Iterative reduction methods⁵²⁾, such as Progressive Meshes⁵³⁾, were widely studied, allowing for gradual refinement or simplification of the model^{54–56)}. In recent years, the trend has shifted towards leveraging advanced computational techniques, such as Parallel Computing^{57–59)} and Machine Learning^{60–62)}, to achieve more efficient and accurate simplification results.

Among the many approaches mentioned above, the QEM algorithm is a commonly used method for mesh simplification, which inspired the proposal in this paper. Michael Garland and Paul S. Heckbert introduced it in 1997¹⁰⁾, then derived many variable species in the following several years^{48–50)63)}.

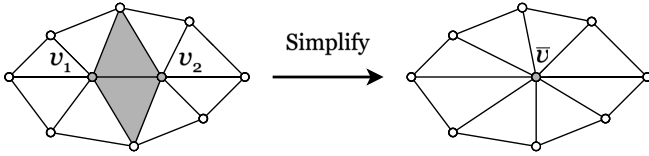


Fig. 6 A typical edge collapse step in QEM Algorithm

As shown in Fig.6, with high-quality model simplification processing, the QEM algorithm minimizes the quadratic error to select and merge vertex pairs efficiently. This merging procedure can also be called edge collapse. QEM algorithm uses Euclidean distance to measure the collapse cost. Specifically, the sum of the squared distances of all neighboring planes $p \in \text{plane}(v_i)$ from the target position $\bar{v}$ to the vertex pair (v_1, v_2) being merged:

$$\bar{v} = \arg \min_v \sum_{p \in \text{plane}(v_1) \cup \text{plane}(v_2)} \text{distance}(v, p)^2 \quad (4)$$

For the merger $(v_1, v_2) \rightarrow \bar{v}$ of any given vertex pair (v_1, v_2) , the quadratic error $\Delta(\bar{v}) = \bar{\mathbf{v}}^\top \mathbf{Q} \bar{\mathbf{v}}$ in target place $\bar{\mathbf{v}} = [v_x, v_y, v_z, 1]^\top$ can be defined as an geometric approximation via a symmetric 4×4 matrix $\mathbf{Q}$:

$$\begin{aligned} \Delta(\bar{v}) &= \Delta([v_x, v_y, v_z, 1]^\top) = \sum_{\mathbf{p} \in \text{planes}(\bar{v})} (\mathbf{p}^\top \bar{\mathbf{v}})^2 \\ &= \sum_{\mathbf{p} \in \text{planes}(\bar{v})} \bar{\mathbf{v}}^\top (\mathbf{p} \mathbf{p}^\top) \bar{\mathbf{v}} \\ &= \bar{\mathbf{v}}^\top \left(\sum_{p \in \text{planes}(\bar{v})} \mathbf{K}_p \right) \bar{\mathbf{v}} \\ &= \bar{\mathbf{v}}^\top \mathbf{Q} \bar{\mathbf{v}} \end{aligned} \quad (5)$$

According Eq.(5), since this error $\Delta(\bar{v})$ is a quadratic function, finding the optimal position $\bar{v}$ is transformed into a linear problem: Assuming the quadratic matrix $\mathbf{Q}$ is a positive definite matrix, the extreme value $\bar{\mathbf{v}}$ of position $\bar{v}$ can be find by partial derivative when $\partial \Delta / \partial x = \partial \Delta / \partial y = \partial \Delta / \partial z = 0$:

$$\begin{bmatrix} q_{11} & q_{12} & q_{13} & q_{14} \\ q_{12} & q_{22} & q_{23} & q_{24} \\ q_{13} & q_{23} & q_{33} & q_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix} \bar{\mathbf{v}} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad (6)$$

3. Proposed Method

The characteristic of 3D volumetric video inspired the approach in this paper: If an algorithm can consider such movement between adjacent frames as conditions, then this mesh simplification algorithm is more advantageous for dynamic mesh than the traditional existing method. As shown in Fig.7, the Tracked QEM Algorithm is proposed to track this dynamic movement, which can introduce the previous frame as a reference via mesh registration.

Compared with the original QEM algorithm used in current volumetric video coding standard⁵⁾, the proposed Tracked QEM Algorithm is mainly improved from three places: introduced mesh registration to generate the reference for simplification; added inter-frame tracking for edge collapse cost function; and improved optimal search strategy when the quadratic error matrix is irreversible.

These modifications ensure that the structure of the simplified mesh has temporal consistency with the previous frame to avoid geometric structures that are too different from those caused by motion and deformation between adjacent frames. The following part of this chapter will comprehensively describe these three main innovation points in this proposed approach.

3.1 Previous frame reference

As discussed in Section 1, the dynamic mesh in 3D volumetric videos is similar to a frame sequence in 2D

video, which also consists of static mesh frames. As the example shown in Fig.8, these static mesh frames captured with a constant time interval Δt^{64} , specific k -th frame in this sequence can also be viewed as a static mesh $mesh^{(k)}$ in traditional computer graph. the subject movement in a specific period $(k-1)\Delta t \sim k\Delta t$ will be recorded by difference between current frame $mesh^{(k)}$ and previous frame $mesh^{(k-1)}$.

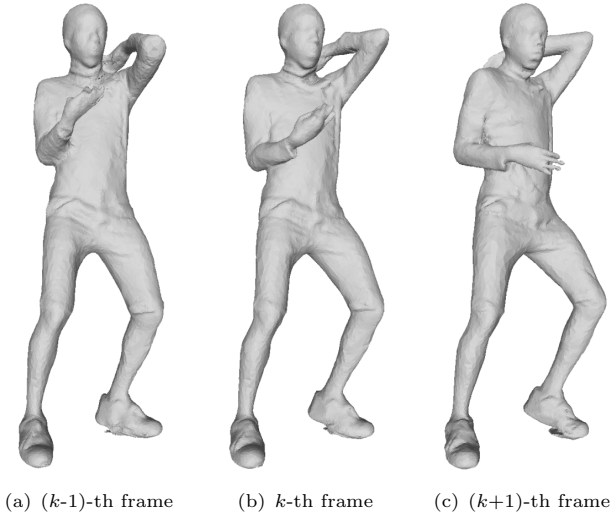


Fig. 8 A frame sequence from dynamic mesh in 3D volumetric video fragment

In order to make the simplified results leverage these differences between two adjacent frames, Tracked QEM introduced the information of the previous frame as a reference in the simplification process. However, due to the inter-frame information, including subject movement, there is no significant correspondence between the two frames in the geometric structure of the static mesh. For this reason, mesh registration was introduced

to track this inter-frame association.

While the 3D volumetric video focuses on the nuances of posture alterations within the volumetric space, it is equally crucial to consider the displacing of the character's position that may occur⁶⁵. This displacement and deformation can be regarded as a combination of rigid and non-rigid transformation¹⁴, which underscores the complexity of capturing and replicating human motion within volumetric environments.

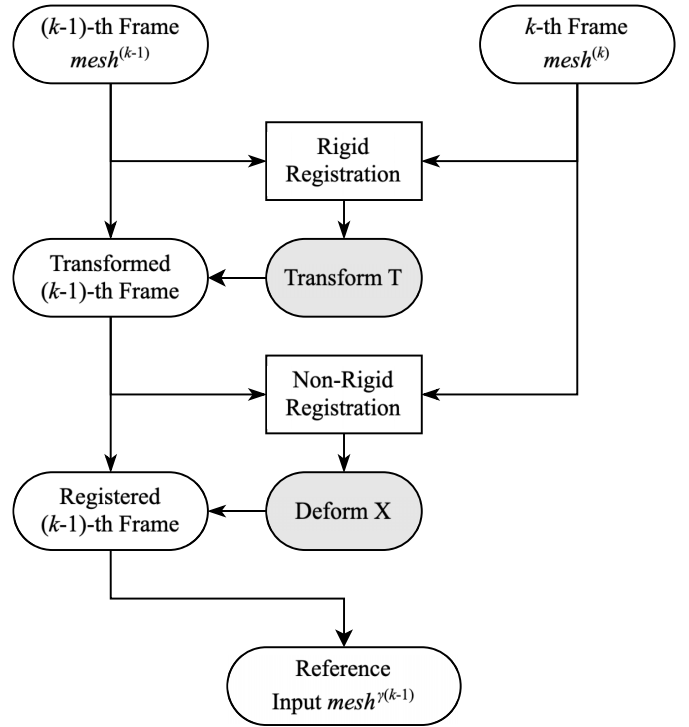


Fig. 9 Mesh registration for reference input to align the $(k-1)$ -th frame

As the flowchart shown in Fig.9, rigid and non-rigid registration is used in the proposed method simultane-

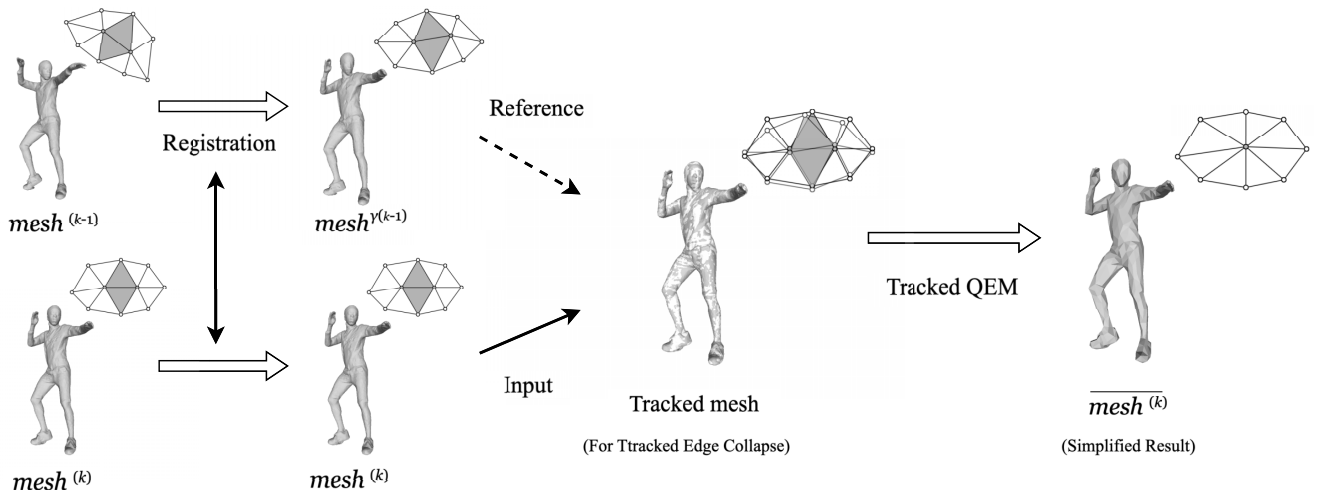


Fig. 7 Flowchart of proposed Tracked QEM Algorithm, including tracked reference from previous frame

ously to track complex human motion¹³). The registered $mesh^{\gamma(k-1)}$ can be obtained by registration $\gamma(\cdot)$ from previous $mesh^{(k-1)}$, which can be used as a reference of current $mesh^{(k)}$ in tracked edge collapse:

1. **Rigid registration:** Rigid registration aims to align inputs to eliminate the possible difficulties caused by displacement in the subsequent processing. The algorithm employed here is the naive Iterative Closest-Point (ICP)²⁶. The previous $mesh^{(k-1)}$ will be moved to align with the current $mesh^{(k)}$ to fetch rigid transformation. The optimization target $\mathbf{T}^*$ in Eq.(1) is changed into the specific format $\mathbf{T}^{\gamma(k-1)}$ in Eq.(7) below, which minimum the corresponding point pair transformation $\mathbf{P}_{ij}^{\gamma(k-1)}$. The result will be passed as input for the subsequent non-rigid registration processing.

$$\mathbf{T}^{\gamma(k-1)} = \arg \min_{\mathbf{T}} \sum_c \left(\mathbf{P}_{ij}^{\gamma(k-1)} \right)^T \Omega_{ij} \left(\mathbf{P}_{ij}^{\gamma(k-1)} \right), \quad (7)$$

$$\mathbf{P}_{ij}^{\gamma(k-1)} = \mathbf{p}_i^{(k)} - \mathbf{T} \oplus \mathbf{p}_j^{(k-1)}$$

2. **Non-rigid registration:** non-rigid deformations will be corrected after rigid alignment. Non-rigid deformations can potentially be in-homogeneous and may occur only in a local region. Therefore, the Fast-RNRR³⁸ based on iteration was selected to obtain acceptable correspondence. Through this Quasi-Newton's solver, the deformation field $\mathbf{X}^*$ in Eq.(2) will be able to be solved by a two-layer iteration. In the proposed method, the affine $\mathbf{R}_j^\gamma$ and displace $\mathbf{t}_j^\gamma$ transformation can be described as $\mathbf{X}^{\gamma(k-1)}$ by the formula in Eq.(8).

$$\mathbf{X}^{\gamma(k-1)} = \min_{\mathbf{X}} E_{\text{align}}(\mathbf{X}) + \alpha E_{\text{reg}}(\mathbf{X}) + \beta E_{\text{rot}}(\mathbf{X}), \quad (8)$$

$$\mathbf{X}^{\gamma(k-1)} = \left\{ \mathbf{X}_j^\gamma \mid \mathbf{X}_j^\gamma = (\mathbf{R}_j^\gamma, \mathbf{t}_j^\gamma) \right\}$$

Consistent with the flowchart shown in Fig.9, once the rigid transformation $\mathbf{T}^{\gamma(k-1)}(\cdot)$ and non-rigid deformation $\mathbf{X}^{\gamma(k-1)}(\cdot)$ has been obtained. Combine the definition from Eq.(2), the registered reference $mesh^{\gamma(k-1)}$ can then be expressed by the transformation matrix $\mathbf{T}^{\gamma(k-1)}$ and the deformation field $\mathbf{X}^{\gamma(k-1)}$:

$$mesh^{\gamma(k-1)} = \gamma_{\text{reg}(k)} \left(mesh^{(k-1)} \right) \quad (9)$$

$$= \mathbf{X}^{\gamma(k-1)} \left(\mathbf{T}^{\gamma(k-1)} \left(mesh^{(k-1)} \right) \right)$$

Mesh registration aims to establish tracking. The correspondence of geometric entities between $mesh^{(k-1)}$ and $mesh^{(k)}$ can obtain this inter-frame tracking. As the reference of current frame $mesh^{(k)}$, related vertices in tracked $mesh^{\gamma(k-1)}$ also need to be introduced into the optimization target $\overline{v^{(k)}}$. It changes the optimization goal from "minimize the effect on current mesh appearance" to "minimize the effect on the previous and current mesh."

3.2 Tracked collapse cost function

As implied by the name, Tracked QEM is based on the original QEM algorithm and has added some new features related to tracking. Consider an edge collapsing on the current k -th frame $mesh^{(k)}$, original QEM contracting two endpoints v_1 and v_2 on a specific edge e to be collapsed into a single point $\bar{v}$. With the same meaning as in Eq.(4), the edge collapse function only considers the faces plane(v_1) and plane(v_2) adjacent collapsed vertices v_1, v_2 from current mesh surface.

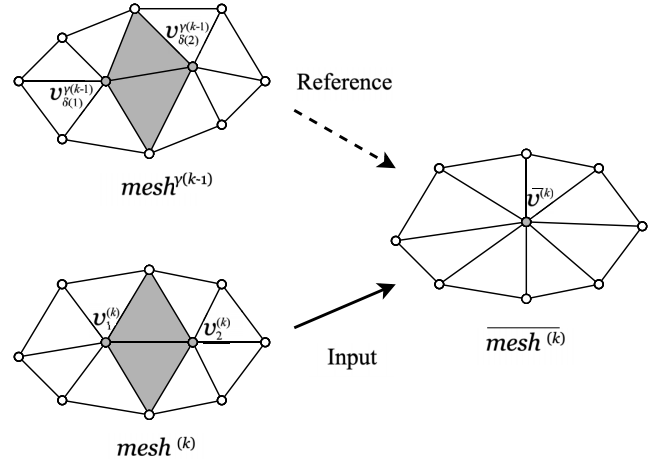


Fig. 10 Tracked Quadric Error Metrics (QEM) Algorithm

Since the inter-frame information was not considered, the original QEM may have led to a lack of temporal consistency in the simplified results. Fortunately, due to the introduction of mesh registration, tracking between two frames was re-established. The Tracked QEM Algorithm redesigned the collapse cost function to utilize this correspondence, as described in Fig.6, above Fig.10 illustrates the proposed design.

Considering the convenience in computing, the similarity between the registration results $mesh^{\gamma(k-1)}$ and the current model $mesh^{(k)}$ should be considered⁴⁰. Which means that when calculating the collapses of edge $e^{(k)}$, optimized items of introduced tracking

$mesh^{\gamma(k-1)}$ in the target $\overline{v^{(k)}}$ should have similar format with $mesh^{(k)}$.

Except for the two vertices $v_1^{(k)}$ and $v_2^{(k)}$ of edge $e^{(k)}$, vertices near to them from tracked $mesh^{\gamma(k-1)}$ will also be considered. Then the optimization target $\bar{v}$ from Eq.(5) changes to $\overline{v^{(k)}}$, where the position has a minimum distance to the adjacent triangle planes $p \in \text{plane}(v_i)$. But except merged $v_1^{(k)}$ and $v_2^{(k)}$ in $mesh^{(k)}$, closed vertices $V_{\delta(i)}^{\gamma(k-1)}$ from $mesh^{\gamma(k-1)}$ in a certain neighbourhood $\delta(i)$ centered around $v_i^{(k)}$ also should be considered:

$$\begin{aligned} \overline{v^{(k)}} &= \arg \min_v \sum_{p \in \text{plane}(v_i)} \text{distance}(v, p)^2, \\ v_i &\in \{v_1^{(k)}, v_2^{(k)}\} \cup V_{\delta(1)}^{\gamma(k-1)} \cup V_{\delta(2)}^{\gamma(k-1)} \end{aligned} \quad (10)$$

The corresponding rough relationship between the two frames has been established through the mesh registration. The position of $mesh^{\gamma(k-1)}$ and $mesh^{(k)}$ are also aligned into the same space coordinates. Hence the vertices $v_{\delta(i)}^{\gamma(k-1)}$ in the neighbourhood $\delta(i)$ constructed collection $V_{\delta(i)}^{\gamma(k-1)}$ from $mesh^{\gamma(k-1)}$ for a specific centre $v_i^{(k)}$ in $mesh^{(k)}$ can be efficiently built through the K-Nearest Neighbor (K-NN) algorithm. Each neighborhood have a specific search radius $\|e^{(k)}\|$, which means the length of edge $e^{(k)} = (v_1^{(k)}, v_2^{(k)})$:

$$\begin{aligned} V_{\delta(i)}^{\gamma(k-1)} &= \left\{ v_{\delta(i)}^{\gamma(k-1)} \mid \|v_{\delta(i)}^{\gamma(k-1)} - v_i^{(k)}\| < \|e^{(k)}\| \right\}, \\ v_{\delta(i)}^{\gamma(k-1)} &\in mesh^{\gamma(k-1)}, v_i^{(k)} \in mesh^{(k)} \end{aligned} \quad (11)$$

According to the QEM matrix in Eq.(5), for the given edge collapse $(v_1, v_2) \rightarrow \bar{v}$, the sum distance $\Delta(v) = \mathbf{v}^\top \mathbf{Q} \mathbf{v}$ between collapsed vertex $\bar{v}$ and surrounding

planes $p \in \text{plane}(v_1) \cup \text{plane}(v_2)$ can be represent by sum of $\mathbf{Q}_1$ and $\mathbf{Q}_2$. Similarly, the tracked quadric error matrix $\mathbf{Q}^{(k)}$ with certain k -th frame $mesh^{(k)}$ can be obtained:

$$\begin{aligned} \mathbf{Q}^{(k)} &= \mathbf{Q}_{\text{track}(1)}^{(k)} + \mathbf{Q}_{\text{track}(2)}^{(k)} \\ &= \mathbf{Q}_1^{(k)} + \mathbf{Q}_{\delta(1)}^{\gamma(k-1)} + \mathbf{Q}_2^{(k)} + \mathbf{Q}_{\delta(2)}^{\gamma(k-1)} \end{aligned} \quad (12)$$

The above formula introduced additional quadric matrices $\mathbf{Q}_{\delta(i)}^{\gamma(k-1)}$, which constructed by the neighbour vertices collection $V_{\delta(i)}^{\gamma(k-1)}$:

$$\mathbf{Q}_{\delta(i)}^{\gamma(k-1)} = \sum_{v \in V_{\delta(i)}^{\gamma(k-1)}} \mathbf{Q}_v^{\gamma(k-1)} \quad (13)$$

It means that when searching for nearby points, the maximum radius of the search will be limited to the side length of $e^{(k)}$, namely $\text{length}(e^{(k)}) = \sqrt{(v_1^{(k)} - v_2^{(k)})^2}$. This restrictive strategy avoids introducing excessive tracking points, resulting in an imbalance in the weight of mesh data between the two frames. The Fig.11 shows the construction process of the neighbour vertices set $V_{\delta(1)}^{(k-1)}$ and $V_{\delta(2)}^{(k-1)}$.

3.3 Approximate optimal search strategy

Another improvement is the search strategy for the approximate optimal collapse position. In the original QEM algorithm, the location of the optimal point $\bar{v}$ is related to the nature of the quadratic matrix $\mathbf{Q}$, which is also the $\mathbf{Q}^{(k)}$ in tracked QEM. According to Eq.(5), suppose the expression of any plane p is $ax + by + cz + d = 0$, the quadratic matrix $\mathbf{Q}_{\text{track}(i)}^{(k)}$ of certain vertex $v_i^{(k)}$ in Eq.(12) can be denoted as the format in Eq.(14).

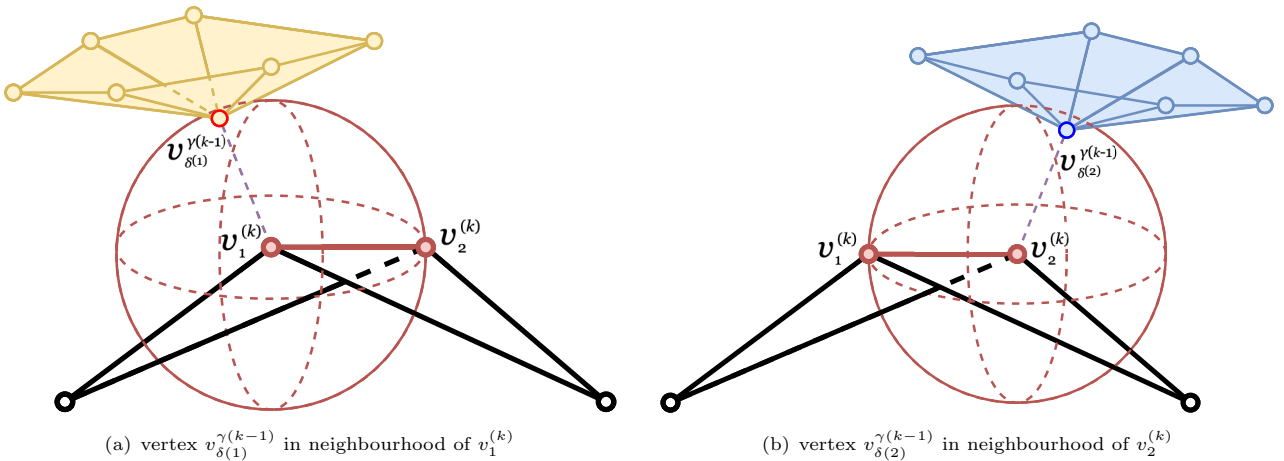


Fig. 11 Nearest Point Search for Edge in Tracked QEM Based on K-NN Algorithm

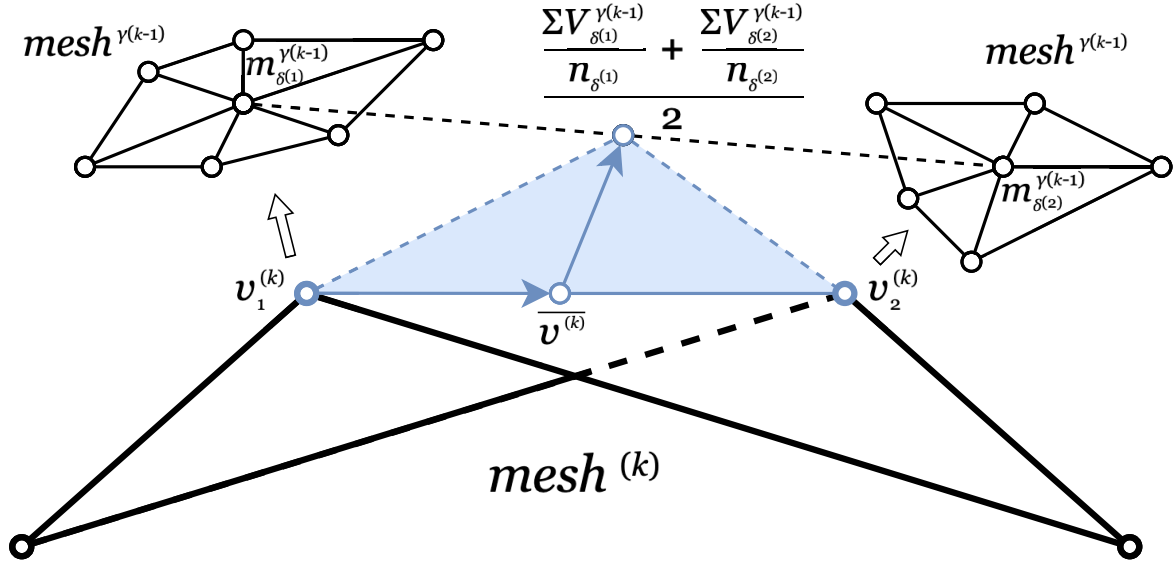


Fig. 12 Approximate solution search on the triangular surface

$$\begin{aligned}
 \mathbf{Q}_{\text{track}(i)}^{(k)} &= \mathbf{Q}_i^{(k)} + \mathbf{Q}_{\delta(i)}^{\gamma(k-1)} \\
 &= \sum_{v \in V_{\delta(i)}^{\gamma(k-1)} \cup \{v_i^{(k)}\}} \sum_{p \in \text{plane}(v)} \mathbf{K}_p \\
 &= \sum_p \begin{bmatrix} a_p^2 & a_p b_p & a_p c_p & a_p d_p \\ a_p b_p & b_p^2 & b_p c_p & b_p d_p \\ a_p c_p & b_p c_p & c_p^2 & c_p d_p \\ a_p d_p & b_p d_p & c_p d_p & d_p^2 \end{bmatrix} \\
 &= \begin{bmatrix} q_{11} & q_{12} & q_{13} & q_{14} \\ q_{12} & q_{22} & q_{23} & q_{24} \\ q_{13} & q_{23} & q_{33} & q_{34} \\ q_{14} & q_{24} & q_{34} & q_{44} \end{bmatrix} \quad (14)
 \end{aligned}$$

As same as the original QEM, according to Eq.(6), the optimized $\bar{v}^{(k)}$ can be solved by the inverse matrix in the form of Eq.(15), which can get the element value through the matrix $\mathbf{Q}^{(k)}$ stated in Eq.(12) calculate by $\mathbf{Q}_{\text{track}(i)}^{(k)}$ in Eq.(14). However, if the quadric error matrix $\mathbf{Q}$ is irreversible, it is necessary to specify the search method to find approximately optimal solutions.

$$\bar{\mathbf{v}} = \begin{bmatrix} q_{11} & q_{12} & q_{13} & q_{14} \\ q_{12} & q_{22} & q_{23} & q_{24} \\ q_{13} & q_{23} & q_{33} & q_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad (15)$$

In reality, only a simple search strategy is needed since the matrix $\mathbf{Q}$ is invertible in most cases. The original QEM algorithm chooses the midpoint position $(v_1 + v_2)/2$ or executes the recursive binary search along

the edge e . But according to Eq.(14), accounted neighbour vertices $V_{\delta(i)}^{\gamma(k-1)}$ in tracked QEM increases the probability that the $\mathbf{Q}_{\text{track}(i)}^{(k)}$ is irreversible. For this reason, more precise search strategies must be designed to ensure the quality of the appearance of the simplified model.

In the original QEM, the search area is the segment defined on edge $e^{(k)}$, which is also restricted on an original mesh surface. Since in the tracked QEM algorithm, the introduction of reference frame $\text{mesh}^{\gamma(k-1)}$, possible solution should not be limited at the $\text{mesh}^{(k)}$ surface anymore. As shown in Fig.12, the proposed search strategy significantly expands the solution space.

The candidate position in the space of the optimized vertex has been expanded into a triangular plane in 3D space. In this proposal, the new search direction is extended by the barycenter $m_{\delta}^{\gamma(k-1)}$ of tracked neighbour vertices set $V_{\delta(1)}^{\gamma(k-1)}$ and $V_{\delta(2)}^{\gamma(k-1)}$, which defined by the endpoint $v_1^{(k)}, v_2^{(k)}$ in current edge $e^{(k)}$:

$$m_{\delta}^{\gamma(k-1)} = \frac{\sum V_{\delta(1)}^{\gamma(k-1)}}{n_{\delta(1)}} + \frac{\sum V_{\delta(2)}^{\gamma(k-1)}}{n_{\delta(2)}} \quad (16)$$

In order to avoid excessive computing expenses, the optimal search will be split into two stages. The first stage is uniform with the original QEM algorithm. Algorithm only proceed search on segment $(v_1^{(k)}, v_2^{(k)})$ to find the optimized result $\bar{v}_{\text{1st}}^{(k)}$ of 1st stage. The second stage is the search on the segment $(\bar{v}_{\text{1st}}^{(k)}, m_{\delta}^{\gamma(k-1)})$. Therefore, the entire progress to get the final $\bar{v}^{(k)}$ can be considered in staged search on the two line segments:

$$(v_1^{(k)}, v_2^{(k)}) \text{ and } (\overline{v_{1st}^{(k)}}, m_\delta^{\gamma(k-1)}).$$

Algorithm 1 Iteration-based binary search

Require: binary search on segment (v_1, v_2)

Ensure: Approximate $\bar{v}$ with smallest cost

Initialization: Iteration count $c \leftarrow 0$

$m \leftarrow \text{middle}(v_1, v_2)$

while $c < C_{max}$ **do**

if $\text{cost}(m) < \min(\text{cost}(v_1), \text{cost}(v_2))$ **then**

$m_1 \leftarrow \text{middle}(v_1, m)$

$m_2 \leftarrow \text{middle}(m, v_2)$

if $\text{cost}(m_1) < \text{cost}(m_2)$ **then**

$v_2 \leftarrow m, m \leftarrow m_1$

else

$v_1 \leftarrow m, m \leftarrow m_2$

end if

end if

 Update: $\bar{v} \leftarrow \min(\text{cost}(v_1), \text{cost}(v_2), \text{cost}(m))$

end while

Ensure: Approximate $\bar{v}$ with smallest cost

As described in Algorithm 1, the search for both stages uses an iteration-based binary search for performance considerations. The algorithm will first select the middle point m of the search line segment to calculate the cost. If the endpoints are less than both sides, two new middle points m_1 and m_2 will be taken on the two new line segments divided by the current middle point. The segment with the middle point with a more negligible cost will be selected for the next iteration. The maximum number of iterations selected in this item is 4. It is worth noting that when the number of iterations is only 1, the algorithm will be degraded to try to use the centre point of triangle $v_1^{(k)}, v_2^{(k)}, m_\delta^{\gamma(k-1)}$ as the approximate solution.

4. Experiment

4.1 Data and Metrics

In the experiment part, the proposed Tracked QEM was compared with the Original QEM algorithm to evaluate the quality and efficiency of the dynamic mesh decimation task. Both algorithms use the same data and conditions for evaluation to ensure the objectivity of the comparison. A few short 3D volumetric video frame sequences are intercepted as the input. The video totalled 150 frames and recorded a human movement. A total of five frame sequences were cut. Each sequence contains five frames. Each frame has about 20,000 valid vertices, 40,000 triangles, and 60,000 edges.

Due to the tracking QEM algorithm always requiring the previous $(k-1)$ -th frame as the additional reference for each simplified k -th frame, each group conducted four experiments. The algorithm will simplify the mesh

from the 2-nd to the 5-th frame, without the 1-st frame. For the same reason, the evaluation will also include the error between each simplified k -th frame and the registered $(k-1)$ -th frame. Simplified results will be visualized and evaluated to analyze the performance of time consistency.

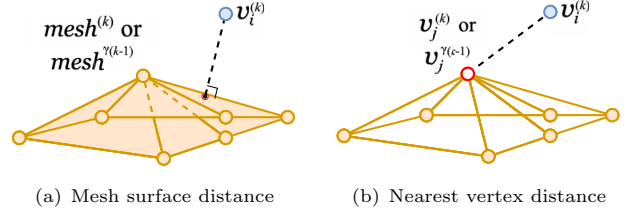


Fig. 13 Indicators based on distance measurement

The evaluation metric of error is designed to measure the inter-frame consistency of the dynamic mesh decimation task in 3D video compression⁶⁶. Inspired by Cloud-Mesh and Cloud-Cloud distance⁶⁷, as shown in Fig.13, mesh surface distance and nearest vertex distance were used as main metrics:

1. Mesh surface distance: The shortest distance from each vertex $\overline{v_i^{(k)}}$ in simplified $\overline{mesh^{(k)}}$ to any point on surface of compared original $mesh^{(k)}$ or referenced $mesh^{\gamma(k-1)}$. Which can be used to measure appearance similarity. The smaller value means the result is closer to the input.

2. Nearest vertex distance: The shortest distance from each vertex $\overline{v_i^{(k)}}$ in simplified $\overline{mesh^{(k)}}$ to the corresponding nearest vertex $v_j^{(k)}$ or $v_j^{\gamma(k-1)}$. Which can be used to evaluate the consistency of geometric topology. The smaller value means more approximate to the input.

4.2 Visualized result

Fig.14 shows the visualized results from the Tracked QEM algorithm. A typical simplified frame with good visual effects was selected and visualized. Fig.14.(a) and Fig.14.(b) is the current frame input $mesh^{(k)}$ and previous frame reference $mesh^{(k-1)}$. It is easy to see from Fig.14.(c) that aligned the input $mesh^{(k)}$ from the current k -th frame and tracked $mesh^{\gamma(k-1)}$ from $(k-1)$ -th frame have a high degree of overlapping in appearance. However, the divergence of local details and geometric topology between the two frames still leads to subtle differences in some areas in the registered result. In traditional mesh decimate processing, this difference will affect the consistency of the simplified frame sequence. This conclusion is why the registered reference $mesh^{\gamma(k-1)}$ should be introduced to Tracked QEM.

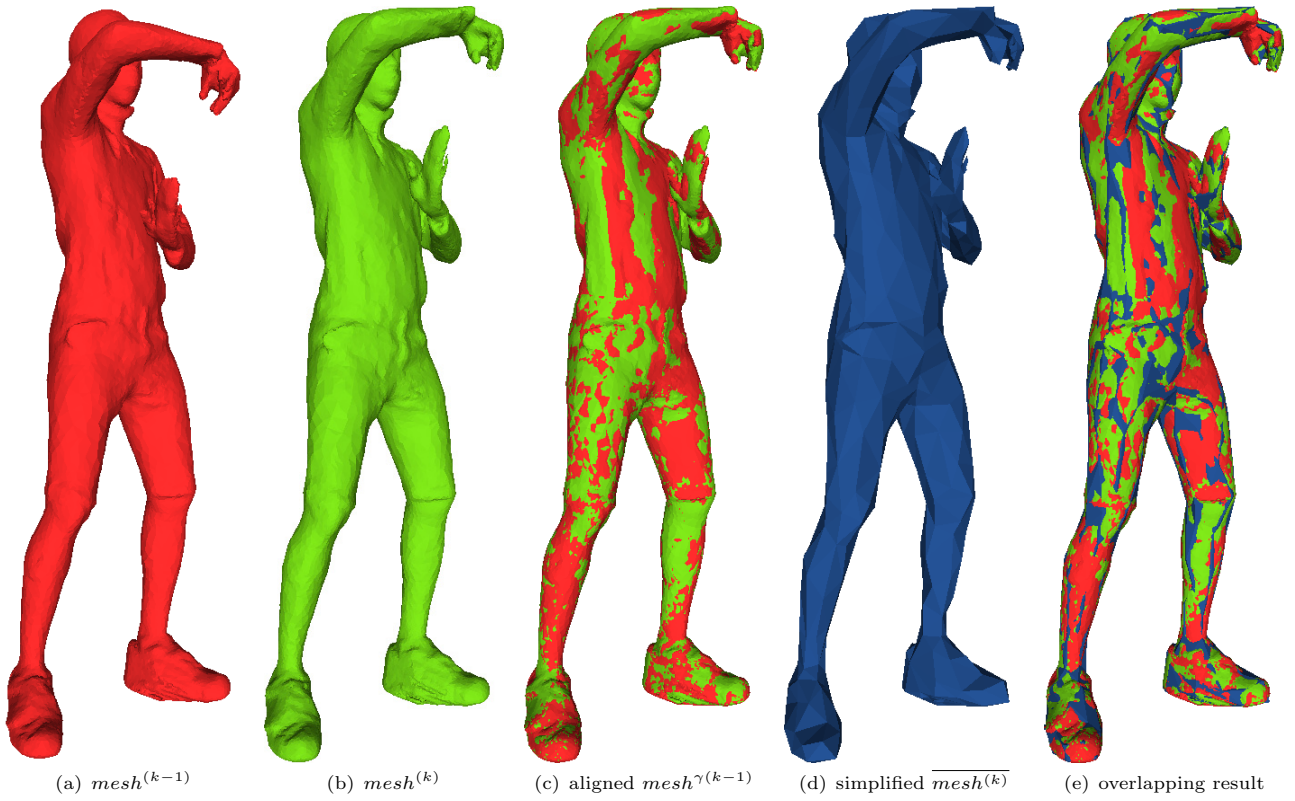


Fig. 14 Visualized inputs & simplification result from Tracked QEM Algorithm

In the Tracked QEM Algorithm, both the spatial information in two input $mesh^{\gamma(k-1)}$ and $mesh^{(k)}$ will be leveraged moderately. These highly similar tracked surface information can be used as a good reference part $V_{\delta(i)}^{\gamma(k-1)}$ in Eq.(10). As shown in Fig.10, the new vertices in simplified result will be generated between $mesh^{\gamma(k-1)}$ and $mesh^{(k)}$, which means convert the result $\overline{mesh^{(k)}}$ shown in Fig.14.(d) from single frame to the inter-frame. It can be observed in Fig.14.(e) that if the alignment results are to be superimposed on the final simplified results, the appearance of both frame

mesh was retained well in most regions.

The additional visualized results from the same frame group support this conclusion by comparing the two input meshes with the simplified results separately. As shown in Fig.15.(a) and Fig.15.(b), respectively superimposed with the simplified results $\overline{mesh^{(k)}}$ using $mesh^{\gamma(k-1)}$ and $mesh^{(k)}$, the areas where they overlap are almost identical. The visualized result shows that the proposed method can also ensure that the simplified results have better temporal consistency and spatial smoothing on geometric structure, even if the movement dramatically changes the attitude of the tracked object between two frames. This result is consistent with the expectations in the Tracked QEM algorithm and the results from the next Comparative experiment.

4.3 Comparative experiment

Consistent with the intuitive feelings in visualized results, introducing the reference frame will make the simplified result also numerically closer to the average of two input $mesh^{\gamma(k-1)}$ and $mesh^{(k)}$ than the traditional decimation method, which can be approved by the stochastic comparison on Mesh surface distance and Nearest vertex distance mentioned before.

Considering the simplified result still has several hundreds or thousands of vertices, the measurement of two distances needs to be summarized using statisti-

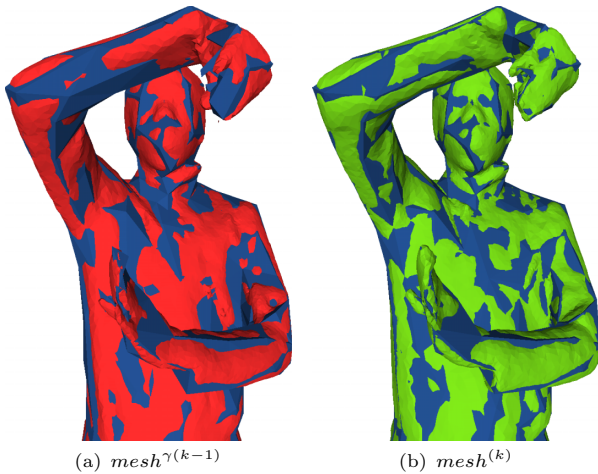


Fig. 15 Comparison of inputs between two frames with simplified results

cal metrics. Five metrics were chosen to quantify algorithm performance in this study: average, Mean, Variance, Standard Deviation, Mean Square Error, and Root Mean Square Error. These metrics are used to characterize the distribution of error distances. Since smaller distance values are usually better, smaller statistical values also mean better results.

Table 1 Best result of mesh surface distance

Algorithm	Mesh	Statistical indicator				
		<i>Ave</i>	<i>Var</i>	<i>STD</i>	<i>MSE</i>	<i>RMSE</i>
Orig QEM	($k-1$)-th	82.73	6143	78.37	56.97	7.547
	k -th	46.23	2129	46.14	21.48	4.635
Track QEM	($k-1$)-th	52.36	2448	49.47	22.42	4.734
	k -th	45.80	1975	44.44	20.41	4.518

Retain 4 valid digits except variance.

Table 2 Best result of nearest vertex distance

Algorithm	Mesh	Statistical indicator				
		<i>Ave</i>	<i>Var</i>	<i>STD</i>	<i>MSE</i>	<i>RMSE</i>
Orig QEM	($k-1$)-th	178.2	28798	169.7	122.4	11.06
	k -th	139.0	22320	149.4	117.3	10.82
Track QEM	($k-1$)-th	144.7	24211	155.6	125.2	11.19
	k -th	134.8	20793	144.2	113.4	10.64

Retain 4 valid digits except variance.

The results in Table.1 and Table.2 are the best results that can be obtained on test data separately for the Tracked and Original QEM algorithm. Table.1 shows that the Tracked QEM achieved better surface distance results in all indicators. This result indicates that unless the gap is not large, the Tracked QEM not only can the simplification of the current frame $mesh^{(k)}$ be achieved without losing the original QEM in the best case, but it can also handle the reference information from previous frame $mesh^{\gamma(k-1)}$ better.

According to the result in Table.2, Tracked QEM can also get the closer vertex distance in the finest of circumstances. Interestingly, although the results of the Original QEM do not include the information from ($k-1$)-th frame, the distance distribution is better than the Tracked QEM. Considering that the results of Tracked QEM do have a better mean and variance, this difference should be treated as reasonable. The reason is that the QEM algorithm only considers the distance to the neighbouring surfaces, and the nearest vertex distance was not part of the optimization objective.

Then is the average performance, which comes from the average of all effective experimental results. The average value of the experimental results is shown in Table.3 and Table.4. Although the best result is better in the value of evaluation metrics, the average performance of the two QEM methods is closer to the real situation in the actual application scenario.

The data in Table.3 and Table.4 shows that Tracked QEM and Original QEM have similar performance at the current k -th frame both on the mesh surface distance and nearest vertex distance. Even on the indicator of mean value, the Tracked QEM algorithm produces a slight degradation.

Table 3 Average value of mesh surface distance

Algorithm	Mesh	Statistical indicator				
		<i>Ave</i>	<i>Var</i>	<i>STD</i>	<i>MSE</i>	<i>RMSE</i>
Orig QEM	($k-1$)-th	101.4	18424	135.7	95.34	9.764
	k -th	57.84	4578	67.66	48.40	6.957
Track QEM	($k-1$)-th	64.26	4246	65.16	55.25	7.433
	k -th	60.96	4122	64.20	52.52	7.247

Retain 4 valid digits except variance.

Table 4 Average value of nearest vertex distance

Algorithm	Mesh	Statistical indicator				
		<i>Ave</i>	<i>Var</i>	<i>STD</i>	<i>MSE</i>	<i>RMSE</i>
Orig QEM	($k-1$)-th	222.4	55731	236.0	220.5	14.85
	k -th	174.5	33238	182.3	176.3	13.27
Track QEM	($k-1$)-th	187.5	37425	193.4	167.7	12.95
	k -th	185.0	35124	187.4	165.9	12.88

Retain 4 valid digits except variance.

Nevertheless, it is easy to see that the two distances of the Tracked QEM at the previous ($k-1$)-th frame are optimized well, which can even achieve similar statistic indicators to the current k -th frame. This result shows that when the topology structure between the two frames changes considerably, the Tracked QEM can capture this change well to achieve better temporal consistency than the traditional QEM in the dynamic Mesh decimation task.

4.4 Analysis and discussion

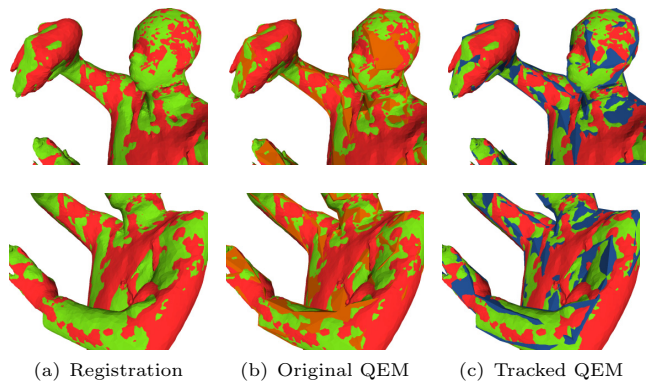


Fig. 16 Comparison with Traditional QEM Algorithm

If the simplified result $mesh^{\gamma(k-1)}$ is aligned to the exact coordinates as the current frame $mesh^{(k)}$, the surfaces of two meshes should be highly coincident and produce some local crossings, as shown in Fig.16.(a). Accordingly, the simplified result $\overline{mesh}^{(k)}$ should be more flattering in appearance to the new overlapped surfaces

after alignment.

Visualization of the comparative experiment can also support conclusions from statistical indicators, result by tracked QEM (Blue mesh in Fig.16.(c)) have a more similar appearance than the original QEM (Orange mesh in Fig.16.(b)). It was evident that the tracked QEM's result is better fitted with finer details like the character's face and palm.

From another perspective, the comparative experiment results show that in most cases, the Tracked QEM algorithm does not significantly improve the simplified performance of the current k -th frame. However, the algorithm's advantage is the temporal consistency between frames; the comparative results prove this opinion.

By introducing reference, Tracked QEM can compress the simplified results from the distance from the registered previous frame $mesh^{\gamma(k-1)}$ to the level similar to the current frame $mesh^{(k)}$, and ensure the simplified $\overline{mesh^{(k)}}$ still have the similar accuracy to the original QEM simultaneously, which proved that the Tracked QEM can significantly improve the time consistency of dynamic mesh decimation without affecting the simplification of the current frame.

For 3D volumetric video compression, mesh decimation is the first stage of the whole encoder pipeline. It aims to reduce the redundant information in the video so that subsequent encoders can compress the video better. The simplification result of track QEM remarkably improves temporal consistency. It makes the simplified mesh sequence more continuous to delineate such redundancies, which helps compress the 3D volumetric video better in the subsequent steps of the encoder pipeline.

5. Conclusion

In this paper, a specialized Tracked QEM Algorithm was proposed, which can use an additional input as a reference from the previous frame to improve temporal consistency between continuous frames. This exploration discussed the potential to extend the conventional mesh simplification algorithm to the dynamic mesh decimation task, offering a tailored solution for mesh decimation in the pre-processing of 3D volumetric video encoding, which seamlessly bridges the difference between consecutive frames.

The most innovative aspect is the application of 3D mesh registration, which facilitates the tracking of mesh models across successive frames, thereby ensuring the

temporal consistency of the simplification process. Due to this improvement, the redesigned collapse cost function and the approximate optimal search strategy guarantee the smoothness of simplified mesh sequences.

The experimental results show that tracked QEM emphasizes its excellent performance in maintaining temporal consistency. The error distance to the registered previous frame is reduced to a close approximation to the current frame. This study demonstrated the potential to improve 3D video compression rates at the pre-processing stage. In future work, the method is expected to be integrated into the standard 3D video coding pipeline to evaluate its impact on compression rates further and inspire subsequent research.

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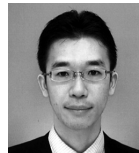
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